PHASE 2 OF 8

Requirement Analysis

Defining what LegalEase must do and what it is built with

Project: LegalEase – AI-Powered Legal Document Generator

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2.1 Purpose of this Phase

The Requirement Analysis phase converts the ideas from brainstorming into clear, testable statements of what LegalEase must do (functional requirements), what it must be built with (technical requirements) and how well it must perform (quality expectations). These requirements act as the reference for design, development and testing.

2.2 Stakeholders and Users

- **End users** who need a legal document draft quickly and want to preview, edit and download it.
- **Project team** (Ganesh Prabhu, Gopinath, Hariharan, Gowtham) who design, build and test the application.
- **Evaluators / reviewers** who assess the completed workflow during the demonstration.

2.3 Functional Requirements

ID	Requirement	Description
FR-1	Select document type	The user can choose the kind of document, e.g. agreement, contract, NDA, lease agreement or employment contract.
FR-2	Enter parties involved	The user can enter the names of the parties to the document.
FR-3	Enter terms and conditions	The user can provide the terms, conditions and any special clauses to be included.
FR-4	Enter effective date and jurisdiction	The user supplies the date from which the document applies and the governing jurisdiction.
FR-5	Generate content	The system sends the inputs to Gemini and receives structured legal-document text.
FR-6	Preview and edit	The generated document is displayed and can be edited by the user before export.
FR-7	Download	The document can be downloaded as TXT, DOCX or PDF.
FR-8	Branding (optional)	Exported documents can include headings, a logo and footer information.

2.4 Input Data Specification

Input field	Type	Required	Example
Document type	Selection	Yes	Lease Agreement
Parties involved	Text	Yes	Party A and Party B names
Terms and conditions	Multi-line text	Yes	Duration, payment terms, obligations
Effective date	Date	Yes	Start date of the agreement
Jurisdiction	Text	Yes	Governing state or country

2.5 Technical Requirements

Component	Purpose in LegalEase
Python	Primary programming language for frontend, backend and formatting logic
Streamlit	Web interface for inputs, preview, editing and download buttons
FastAPI and Uvicorn	Backend REST API (/generate endpoint) and the server that runs it
Google Gemini Generative AI	Generates the structured legal-document text
python-docx	Builds formatted Word (.docx) documents
FPDF	Builds formatted PDF documents
Pillow	Handles image processing for logos in exported files
Requests and python-dotenv	Frontend-to-backend HTTP calls; loading the API key and settings from an environment file

2.6 Non-Functional Requirements

- **Usability:** a simple single-page form that needs no technical knowledge.
- **Performance:** the draft should be returned within a reasonable waiting time, subject to Gemini response speed, with a visible progress indicator.
- **Reliability:** invalid input or AI/API failure must produce a clear message instead of a crash.
- **Security:** the Gemini API key is kept in an environment file and is never hard-coded or shown to the user.

- **Maintainability:** frontend, backend, AI logic and formatters are separate modules so each can be changed independently.
- **Portability:** the application runs locally on a standard Python environment.

2.7 User Stories

1. As a user, I want to select a document type so that the draft matches my need.
2. As a user, I want to enter parties, terms, date and jurisdiction so that the draft is specific to my situation.
3. As a user, I want to edit the generated text so that I can correct or improve it.
4. As a user, I want to download the draft as DOCX or PDF so that I can share or print it.

2.8 Assumptions and Constraints

- The user has an internet connection and a valid Gemini API key is configured.
- Generated documents are drafts and are not a substitute for professional legal advice.
- The application is intended for local deployment and demonstration.
- Only the document types and export formats listed above are in scope.

2.9 Acceptance Criteria

The requirements are considered met when a user can enter all required inputs, obtain a generated draft, edit it, and successfully download readable TXT, formatted DOCX and formatted PDF files, with invalid requests rejected by the API. Each requirement is later verified by a matching test case in the Testing phase.

Traceability: FR-1 to FR-4 are verified by the input test, FR-5 by the generation and API tests, FR-6 by the preview and edit tests, and FR-7 by the three export tests.

2.10 Detailed Use Case: Generate a Document

Element	Description
Actor	End user
Precondition	Backend and frontend are running; Gemini API key is configured
Main flow	Select document type; enter parties, terms, effective date and jurisdiction; click Generate Document; review the preview; edit if required; download in the chosen format
Alternate flow	If a required field is empty, the request is rejected and the user is asked to complete it
Exception flow	If Gemini is unreachable, an error message is shown and the user may retry

Element	Description
Postcondition	A draft is displayed and available for download as TXT, DOCX or PDF

2.11 Supported Document Types

The application is designed around the categories identified during brainstorming. Because content is generated by AI rather than filled into fixed templates, the same input form can serve all of them.

- Agreements and contracts (general purpose)
- Non-disclosure agreements (NDAs)
- Lease agreements
- Employment documents such as employment contracts

2.12 Output Requirements

Output	Requirement
Preview	Full generated text shown in an editable area with no truncation
TXT	Readable plain text file containing the final (edited) content
DOCX	Word file with headings, structured clauses and optional logo and footer
PDF	PDF file with the same structure, readable on any device

2.13 Requirement Priorities

Requirements FR-1 to FR-7 are mandatory for a complete demonstration. FR-8 (branding) is desirable and is implemented after the core workflow is stable. Non-functional requirements on reliability and security apply to every functional requirement.

2.14 Out of Scope

- User accounts, login and saved-document history
- Electronic signatures
- Automatic legal validation or certification of the generated content
- Cloud deployment beyond local demonstration